

Mutual Information

math-foundations / concept 39

1 Setting

Let X and Y be random variables on a common probability space, taking values in finite (or countable) alphabets $\mathcal{X}$ and $\mathcal{Y}$. Write $p(x, y)$ for the joint probability mass function and $p(x), p(y)$ for the marginals. All logarithms are natural unless otherwise stated; replace $\log$ with $\log_2$ to work in bits.

2 Definitions

Definition 1 (Mutual information, KL form). *The mutual information of X and Y is*

$$I(X; Y) := D_{\text{KL}}(p(x, y) \| p(x)p(y)) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log \frac{p(x, y)}{p(x)p(y)},$$

with the conventions $0 \log 0 = 0$ and $0 \log(0/0) = 0$.

Definition 2 (Entropic form). *Equivalently,*

$$I(X; Y) = H(X) - H(X | Y) = H(Y) - H(Y | X) = H(X) + H(Y) - H(X, Y),$$

where $H(X) = -\sum_x p(x) \log p(x)$ and $H(X | Y) = -\sum_{x,y} p(x, y) \log p(x | y)$.

3 Main theorem

Theorem 1 (Nonnegativity of mutual information). *For all random variables X, Y ,*

$$I(X; Y) \geq 0,$$

with equality if and only if X and Y are independent.

Proof. By definition, $I(X; Y) = D_{\text{KL}}(p_{XY} \| p_X p_Y)$, where $p_{XY}(x, y) = p(x, y)$ and $(p_X p_Y)(x, y) = p(x)p(y)$. Both functions are probability mass functions on $\mathcal{X} \times \mathcal{Y}$: the first by construction, and the second because $\sum_{x,y} p(x)p(y) = (\sum_x p(x))(\sum_y p(y)) = 1$.

From the KL-divergence node (concept 38, Gibbs' inequality), for any two pmfs p, q on a common alphabet $\mathcal{Z}$,

$$D_{\text{KL}}(p \| q) = \sum_z p(z) \log \frac{p(z)}{q(z)} \geq 0, \tag{1}$$

with equality if and only if $p(z) = q(z)$ for every z with $p(z) > 0$ (equivalently, $p = q$ as measures). Applying (1) with $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$, $p = p_{XY}$, and $q = p_X p_Y$ gives $I(X; Y) \geq 0$ at once.

For the equality condition: by Gibbs, $I(X; Y) = 0$ iff $p(x, y) = p(x)p(y)$ for every pair $(x, y) \in \mathcal{X} \times \mathcal{Y}$ — which is precisely the definition of independence of X and Y (concept 21). $\square$

Corollary 1 (Symmetry). $I(X; Y) = I(Y; X)$.

Proof. The KL form is manifestly symmetric: swapping the summation indices leaves both $p(x, y)$ and $p(x)p(y)$ invariant. Equivalently, $H(X) + H(Y) - H(X, Y)$ is symmetric in X and Y . $\square$

4 Worked example: binary symmetric channel

Let $X \sim \text{Bernoulli}(1/2)$ and $Y = X \oplus N$, with $N \sim \text{Bernoulli}(\varepsilon)$ independent of X , $\varepsilon \in [0, 1]$. Then $Y \sim \text{Bernoulli}(1/2)$, so $H(X) = H(Y) = \log 2$, and $H(Y | X) = H_2(\varepsilon) := -\varepsilon \log \varepsilon - (1 - \varepsilon) \log(1 - \varepsilon)$. Therefore

$$I(X; Y) = \log 2 - H_2(\varepsilon).$$

At $\varepsilon \in \{0, 1\}$ the channel is deterministic and $I = \log 2$; at $\varepsilon = 1/2$ the channel is pure noise and $I = 0$ — consistent with Theorem 1, since $\varepsilon = 1/2$ makes $X \perp Y$.